

DSA / Algorithms (Abdul Bari) - Day 43

45-day plan mapped to the Algorithms playlist

KMP String Matching

Knuth-Morris-Pratt pattern matching.

Watch these Abdul Bari videos

- > 9.1 Knuth-Morris-Pratt KMP String Matching Algorithm

Key concepts

- Avoids re-checking characters using a prefix (LPS) table.
- $O(n+m)$ matching.
- Failure function drives skips.

Key idea

LPS table tells how far to shift on a mismatch.

Tasks (1.5h)

- ☐ Watch the video
- ☐ Build the LPS table
- ☐ Code KMP

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0